

Provincial Healthcare Network Architecture

Design Summary | Manthan Ray, Regina, SK | Design project: 1 data centre, 2 hospitals, 5 clinics

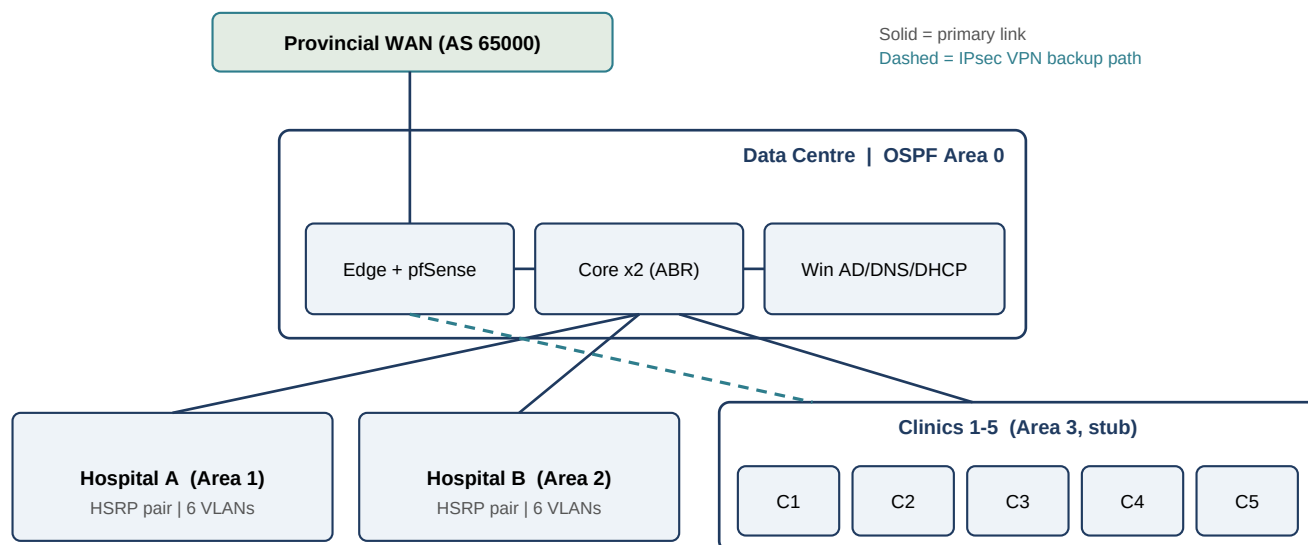
1. Objective

Design a secure, resilient, and scalable network connecting a data centre, two hospitals, and five rural clinics for a fictional provincial health authority. The design supports clinical applications, voice, and Wi-Fi, protects patient information, and can be rolled out in phases with limited downtime.

2. Requirements and Design Response

Requirement	Design response
Clinical systems available 24/7	Dual cores, EtherChannel, HSRP at hospitals, backup IPsec path for every clinic
Patient data protected	Six VLANs, least-privilege ACLs, pfSense edge firewall, IPsec encryption, central logging
Small clinics, limited IT staff	One standard clinic template: same VLANs, policies, and QoS at every site
Reliable voice for care teams	Dedicated VoIP VLAN, DSCP EF (46) marking, priority queuing on WAN links
Room to grow	Hierarchical design, OSPF area summarization, capacity for additional clinics
Manageable at scale	Zabbix monitoring, SNMPv3, syslog, automated nightly config backups (Python)

3. Architecture



- **Segmentation:** VLAN 10 Clinical, 20 Admin, 30 VoIP, 40 Guest, 50 Medical devices, 99 Management.
- **Addressing:** Data centre 10.10.0.0/16, Hospital A 10.20.0.0/16, Hospital B 10.30.0.0/16, clinics 10.100.N.0/24.
- **Routing:** OSPF with Area 0 at the data centre, Areas 1 and 2 at hospitals, Area 3 (totally stubby) for clinics; eBGP to the provincial WAN provider.
- **Services:** Windows Server (Active Directory, DNS, DHCP with relay) and Ubuntu (Zabbix, rsyslog): a mixed Windows/Unix environment.

4. Key Decision: WAN Approach

Two options were compared with a weighted decision matrix (scores are my judgment, 1 to 5).

Criteria	Weight	A: Private WAN at every site	B: Broadband + IPsec (hybrid)
Reliability	30%	5	3
Security	25%	5	4
Cost	20%	2	4
Scalability	15%	3	4
Time to deploy	10%	2	4
Weighted score		3.8	3.7

Recommendation: a hybrid. Private links for the data centre and hospitals, where downtime affects patient care most. Clinics use a lower-cost primary WAN link with broadband and IPsec as the backup path. Scores were close, so the decision matches spending to clinical risk at each site.

5. Security and Privacy

- Clinical VLAN reaches the clinical application over HTTPS (TCP 443) only; guest VLAN reaches the internet only; medical devices reach only their servers.
- Management by SSH from the management VLAN only, with AAA, SNMPv3, OSPF message-digest authentication, and central syslog.
- IPsec site-to-site VPN using IKEv2 with AES-256, SHA-256, and DH group 14.
- Designed with the principles of Saskatchewan's Health Information Protection Act in mind: need-to-know access, safeguards in transit, and audit logging. This is a design intent, not a compliance claim.
- Recommended next steps: 802.1X network access control and intrusion detection.

6. Validation Plan

Ten test cases cover DHCP relay, VLAN isolation, ACL enforcement, WAN failure and failover, HSRP failover, VPN operation, voice and QoS marking, monitoring alerts, config backup, and management-access restrictions. Each test has defined steps and expected results; the test plan is included in the project repository.

7. Business Case and Rollout

The business case compares three-year total cost (one-time plus three years of recurring cost) and the operational benefits of each option: less downtime, faster clinic onboarding through a standard template, and lower support effort through central monitoring.

Phase	Activities	Exit criteria
1. Design approval	Requirements sign-off, architecture and security review	Approved design
2. Pilot	Deploy one clinic, run the full test plan	All tests pass
3. Hospitals	Staged cutovers in maintenance windows, rollback ready	No critical incidents
4. Remaining clinics	Repeat the clinic template	Each site tested
5. Decommission	Retire old links after validation	Legacy services off

Top risks and mitigations: outage during cutover (off-hours windows, pilot first, rollback plan); misconfigured ACL blocking clinical traffic (lab testing, peer review, staged rollout); clinic staff unfamiliar with the new setup (standard template, short runbook, training).

8. Standards and Tools

Architecture approach: TOGAF-informed (vision, business, information systems, technology, migration planning, governance); not certified.

Tools: GNS3 / Cisco Packet Tracer (partial lab build), Python (Netmiko). The design also covers pfSense, Windows Server, Ubuntu, and Zabbix.

Deliverables: requirements, addressing plan, topology diagrams, sanitized configs, automation script, test plan, rollout plan. Configuration templates, diagrams, and scripts: github.com/manthanray03/healthcare-network-design.